Financial functions in Excel are specialized formulas designed to perform complex calculations involving money, such as loan payments, investment growth, and the time value of money.

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1. Core Time Value of Money (TVM) Functions

These functions help you understand the value of money over a specific period, which is essential for personal and business financial planning.

- **PMT (Payment)**: Calculates the periodic payment for a loan based on constant payments and a constant interest rate.
  - *Usage*: Determining monthly mortgage or car loan payments.
- **FV (Future Value)**: Determines the future value of an investment based on a constant interest rate and regular payments.
  - *Usage*: Estimating how much a retirement fund will be worth after 20 years of monthly contributions.
- **PV (Present Value)**: Calculates the current worth of a future stream of cash flows or a lump sum.
  - *Usage*: Deciding if a future payout is worth more than a specific amount of cash today.

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2. Investment Appraisal Functions

These functions are used to evaluate the profitability of a potential project or investment.

- **NPV (Net Present Value)**: Calculates the net present value of an investment by using a discount rate and a series of future payments and income.
- **IRR (Internal Rate of Return)**: Returns the internal rate of return for a series of cash flows. It is used to estimate the profitability of potential investments.

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3. Loan and Interest Analysis

Beyond basic payments, these functions break down where your money is going.

- **IPMT (Interest Payment)**: Calculates the interest portion of a specific loan payment.
- **PPMT (Principal Payment)**: Calculates the principal portion of a specific loan payment.
- **RATE**: Returns the interest rate per period of an annuity.
- **NPER**: Returns the number of periods for an investment or loan.

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4. Logical and Data Support for Financial Functions

For financial functions to work accurately, the data must be properly formatted and organized:

- **Cell Category**: Financial results should be formatted using the "Currency" or "Accounting" data types to ensure correct decimal placement and symbol usage.
- **Absolute References**: When calculating loan tables, use absolute references (e.g., **B1**) for the interest rate so the formula stays locked when copied down a column.
- **Data Validation**: Restrict input cells (like interest rates or loan terms) to specific ranges to prevent unrealistic calculations.
- **IFERROR()**: Since financial functions can return errors if the math is impossible (like a 0% interest rate in certain formulas), use **IFERROR** to provide a clean message.

Key Rules for Financial Accuracy

- **Consistent Units**: Ensure the interest rate and the number of periods match (e.g., if payments are monthly, divide the annual interest rate by 12).
- **Cash Flow Sign Convention**: In Excel financial functions, money you pay out is typically represented as a **negative number**, and money you receive is a **positive number**.